

Prague University of Economics and Business

Faculty of Informatics and Statistics



**EXPLORING THE VARIANTS OF THE TURING
TEST: EVALUATING ARTIFICIAL
INTELLIGENCE AND ITS RELATIONSHIP TO
HUMAN INTELLIGENCE**

BACHELOR THESIS

Study programme: Applied Informatics

Field of study: Applied Informatics

Author: Aubakir Utenov

Bachelor thesis supervisor: Mgr. Ing. Tomáš Sigmund, Ph.D.

Prague, December 2023

Declaration of Authenticity

I hereby declare that this thesis is my original work. All sources and materials used have been properly acknowledged and referenced. This thesis has not been submitted for any degree or examination at any other university or institution.

In Prague on

.....

Aubakir Utenov

Acknowledgement

I want to express my deepest gratitude to Mgr. Ing. Tomáš Sigmund, Ph.D., my thesis supervisor, for his support and guidance throughout this research. His expertise and feedback have been instrumental in shaping this thesis. I am also immensely grateful to my family for their belief in my abilities and moral support.

Abstract

This thesis delves into the Turing test (TT) and its variants, examining its relevance in the ongoing discourse of artificial and human intelligence.

The primary aim is to assess TT variants, evaluate their effectiveness in measuring artificial intelligence (AI) systems' human-like intelligence, and compare this with human intelligence. This involves three steps: first, analyzing different TT versions, focusing on their reliability and the complexity of AI in replicating human intelligence, including the original TT and its modifications. Second, exploring human intelligence to establish a clear benchmark for AI comparison. Third, comparing similar features of AI tools ChatGPT and Midjourney with each other and human perception using variants of TT. This thesis aims to contribute to the broader discourse on the nature of intelligence by comparing human and machine intelligence.

Methodologically, the thesis is divided into theoretical research and practical analysis. The theoretical part, underpinned by a literature review, involves a conceptual analysis of human intelligence for a comparative framework. Practically, the study evaluates two AI tools, ChatGPT and Midjourney, for their demonstration of verbal and visual intelligence aspects related to the TT. These AI tools are subjected to human-operated tests, assessing their creativity and visual perception against human intelligence criteria.

In conclusion, the imperfection of TT variants in assessing the presence of human behavior and qualities in AI was shown. The findings offer insights into the current state of AI and its potential trajectory, emphasizing the evolving relationship between human and machine intelligence.

Keywords

artificial intelligence, human intelligence, Turing test.

Abstrakt

Tato práce se zabývá Turingovým testem (TT) a jeho variantami, zkoumá jeho relevanci v probíhajícím diskurzu o umělé a lidské inteligenci.

Hlavním cílem je posoudit varianty TT, vyhodnotit jejich účinnost při měření lidské inteligence artificial intelligence (AI) systémů a porovnat to s lidskou inteligencí. To zahrnuje tři kroky: první, analýza různých verzí TT, zaměřená na jejich spolehlivost a složitost AI při replikaci lidské inteligence, včetně původního TT a jeho modifikací. Druhý, prozkoumání lidské inteligence, aby bylo stanoveno jasné měřítko pro srovnání s AI. Třetí, porovnání podobné funkčnosti AI nástrojů ChatGPT a Midjourney mezi sebou a s lidským vnímáním pomocí variant TT. Tato práce si klade za cíl přispět do širšího diskurzu o povaze inteligence porovnáním lidské a strojové inteligence.

Metodologicky je práce rozdělena na teoretický výzkum a praktickou analýzu. Teoretická část, podložená rešerší literatury, zahrnuje konceptuální analýzu lidské inteligence pro komparativní rámec. Prakticky studie hodnotí dvě AI nástroje, ChatGPT a Midjourney, pro jejich demonstraci verbálních a vizuálních aspektů inteligence souvisejících s TT. Tyto AI nástroje jsou podrobeny testům prováděným lidmi, hodnotící jejich kreativitu a vizuální vnímání vůči kritériím lidské inteligence.

Závěrem byla ukázána nedokonalost variant TT při posuzování přítomnosti lidského chování a vlastností v AI. Závěry poskytují náhledy do současného stavu AI a jeho možného směřování, s důrazem na vyvíjející se vztah mezi lidskou a strojovou inteligencí.

Klíčová slova

lidská inteligence, Turingův test, umělá inteligence.

Contents

Introduction.....	10
Aim of the thesis	10
Methodology	11
1 The Original Turing Test and its Variants	12
1.1 Imitation Game.....	12
1.2 Total Turing Test.....	13
1.3 Reverse Turing Test	14
1.4 Minimal Turing Test	15
1.5 Visual Turing Test	17
1.6 Lovelace Test	18
1.7 Comparison	19
2 Relationship of the Turing Test Variants to Human Intelligence	22
2.1 Definition of the Human Intelligence.....	22
2.1.1 Theory of Multiple Intelligences	22
2.1.2 Theory of Successful Intelligence	23
2.1.3 PASS theory of intelligence.....	24
2.2 Imitation Game.....	24
2.3 Total Turing Test	24
2.4 Reverse Turing Test.....	25
2.5 Minimal Turing Test.....	25
2.6 Visual Turing Test.....	26
2.7 Lovelace Test.....	26
2.8 Conclusion	26
3 Evaluation of the Turing Test Concept.....	27
3.1 Chinese Room Argument and its critique of the Turing Test's ability to capture genuine understanding.....	27
3.2 Evaluation of how well the Turing Test can evaluate the similarity of AI to human intelligence.....	29
3.3 Assessment of whether machine intelligence has such qualities as emotional intelligence, intuition, self-awareness, morals and ethics, physical and sensory intelligence.....	30
3.3.1 Emotional intelligence	30
3.3.2 Intuition	31
3.3.3 Self-awareness	31
3.3.4 Morals and Ethics.....	31
3.3.5 Physical intelligence	32
3.4 The Ideal Turing Test	32
3.5 Conclusion	33
4 Practical Applications of AI Tools	34
4.1 ChatGPT	35
4.1.1 Interface and Features of ChatGPT in the Web Version	35
4.1.2 Limitations	38
4.1.3 ChatGPT Visual Turing Test	38

4.2 Midjourney	43
4.2.1 Interface and Features of Midjourney in Discord	44
4.2.2 Limitations	45
4.2.3 Midjourney and ChatGPT Lovelace Test	45
4.3 Conclusion	49
Conclusions	51
List of references	52

A List of Figures

Figure 1: Judgement results from the study (McCoy and Ullman 2018).....	16
Figure 2: The Seven Intelligences (Gardner and Hatch 1989).....	23
Figure 3: ChatGPT web interface (https://chat.openai.com/).....	35
Figure 4: ChatGPT Custom Instructions window (https://chat.openai.com/).....	36
Figure 5: Image 1 (Geman et al. 2015).....	39
Figure 6: Image 2 (Geman et al. 2015).....	41
Figure 7: Example of Midjourney prompt.....	43
Figure 8: Image grid.....	43
Figure 9: Buttons.....	44
Figure 10: Midjourney Task 1 output.....	45
Figure 11: ChatGPT Task 1 output.....	45
Figure 12: Midjourney Task 2 output.....	45
Figure 13: ChatGPT Task 2 output.....	45
Figure 14: Midjourney Task 3 output.....	46
Figure 15: ChatGPT Task 3 output.....	46
Figure 16: Midjourney Task 4 output.....	46
Figure 17: ChatGPT Task 4 output.....	46
Figure 18: Midjourney Task 5 output.....	47
Figure 19: ChatGPT Task 5 output.....	47
Figure 20: Midjourney Task 6 output.....	47
Figure 21: ChatGPT Task 6 output.....	47
Figure 22: Midjourney Task 7 output.....	48
Figure 23: ChatGPT Task 7 output.....	48

List of Abbreviations

AI	Artificial intelligence
CAPTCHA	Completely Automated Public Turing test to tell Computers and Humans Apart
GPT	Generative Pretrained Transformer
LT	Lovelace test
MTT	Minimal Turing test
RTT	Reverse Turing test
TT	Turing test
TTT	Total Turing test
VTT	Visual Turing test

Introduction

In the ever-evolving world of technology, the field of AI is a testament to human ingenuity and the relentless pursuit of understanding intelligence. The philosophical roots of AI research go back centuries and are intertwined with the human desire to understand the essence of consciousness and intelligence. This thesis explored one of the most important milestones in AI research — the TT and its many variants.

The relevance of AI in everyday life has grown from theoretical speculation to practical applications, permeating various industries and changing the way people interact with technology. The TT, proposed by Alan Turing in the mid-20th century, has become a concept that offers a method for assessing machine intelligence by its indistinguishability from human intelligence. However, as AI has evolved, the TT has been subject to criticism and modification, leading to various variations that attempt to assess the capabilities of AI more complex.

His passion for AI tools drove the author to delve into this topic. This work was a personal quest to understand the relationship between human and machine intelligence. Exploring the TT and its variants provided a basis for scrutinizing the evolving relationship between humans and machines, challenging the boundaries of what humans consider intelligent behavior.

Aim of the thesis

The primary aim of this thesis was to conduct an assessment of the different versions of the TT to determine their reliability and effectiveness in evaluating the level of human-like intelligence in AI systems, focusing on comparing these findings directly with human intelligence. This entailed a threefold approach:

1. The thesis's initial part was dedicated to examining the different TT versions. This involved an analysis of each variant's focus, methodology, and criteria. The focus was on the reliability, simplicity, and complexity of assessed aspects of AI in terms of its ability to mimic or replicate human intelligence. This evaluation included the original TT that Alan Turing proposed and its subsequent adaptations and modifications.
2. Following the assessment of TT variants, the thesis explored the concept of human intelligence. This segment aimed to define and describe the various facets of human intelligence, providing a clear benchmark for comparison with AI systems. The intention was to offer a comprehensive understanding of what constitutes human intelligence and evaluate the quality of TT variants in assessing the level of AI from the perspective of human intelligence.

3. The practical part of the thesis compared the AI tools ChatGPT and Midjourney with each other and with human perception using variants of TT: the Visual Turing test (VTT) and the Lovelace test (LT). The purpose of this part was to test the persuasiveness of AI tools in demonstrating visual perception skills and creativity.

Ultimately, this research attempted to contribute to the ongoing AI and cognitive science discourse, offering a perspective on the relationship and comparability between human and machine intelligence.

Methodology

The methodology of this thesis is divided into two main parts: theoretical research and practical analysis.

1. Theoretical research: The theoretical foundation of this thesis was established through the literature review. This involved the study of both classic and contemporary books, academic journals, and online sources relevant to the TT, its variants, and human intelligence. The theoretical part included a conceptual analysis of human intelligence based on the works of reliable authors. It served as a benchmark for comparing the performance of AI tools.
2. Practical analysis:
 - 2.1. For the practical part of the thesis, two popular narrow AI-based tools were analyzed: ChatGPT (a text-based AI tool) and Midjourney (an AI tool with capabilities in visual tasks). These tools have been chosen due to their prevalence and the different types of intelligence they demonstrate, aligning with the verbal and visual aspects of the TT.
 - 2.2. The practical analysis involved conducting specific human-operated tests with ChatGPT and Midjourney, using predefined criteria based on the VTT and LT. The outcomes were analyzed to determine how these AI tools exhibit human-like intelligence, specifically in creativity and visual perception.
 - 2.3. The results from the theoretical research were then used to assess how closely the AI tools can mimic human capabilities.

1 The Original Turing Test and its Variants

“In June 1948, the world’s first electronic stored-program digital computer, the Manchester “Baby”, ran its first program in the Laboratory. During 1948–49, the Baby was expanded to become a much more substantial machine, and in May 1949 the Electronic Delay Storage Automatic Calculator (EDSAC) became the second electronic stored-program computer to function, at the Mathematical Laboratory of the University of Cambridge. Further developments at Manchester led to the first commercially available electronic digital computer, in February 1951. In this way, in the late 1940s and early 1950s, the abstract question whether machine intelligence is possible came to be focused on a particular form of machine. Could machinery like the Baby and the EDSAC, if given additional high-speed memory and enhanced processing speed, be said to think?” (Proudfoot 2013, p. 391)

Alan Turing formulated the Imitation Game concept in his paper, "Computing Machinery and Intelligence" (1950) to explore this question.

1.1 Imitation Game

The Imitation Game primarily addresses the question, "Can machines think?" Instead of directly engaging with this question, Turing proposes a behavior-based approach, implying that if a machine can exhibit intelligent behavior indistinguishable from a human, it would be reasonable to consider that it "thinks."

The focus of the Imitation Game is to see whether machines can exhibit behavior indistinguishable from that of a human. Turing substitutes the initial question, "Can machines think?" with a more operationally defined question, *“Are there imaginable digital computers which would do well in the imitation game?”* (Turing 1950, p. 442).

The Imitation Game involves three participants: a human interrogator, a human respondent, and a machine respondent. The interrogator is in a separate room and communicates with the respondents through typed text. The objective of the interrogator is to determine which respondent is human and which is the machine. The respondents aim to convince the interrogator of their human identity. If the interrogator cannot distinguish between the human and the machine, or if the machine is mistaken for the human as often as the human is, the machine is said to have passed the test.

It's worth noting that the original game proposed by Turing involved a male and a female respondent, where the interrogator's task was to determine their genders. Turing then adapted the game to evaluate machine intelligence (Turing 1950).

Turing suggested that a machine can be considered intelligent when it can imitate a human to the point that an interrogator could not distinguish between them based on their responses to questions. He proposed a criterion for success, stating, if *“an average*

interrogator will not have more than 70 percent chance of making the right identification after five minutes of questioning” (Turing 1950, p. 442).

Turing’s Imitation Game moves the discussion from a philosophical point of view to a more empirical and operational one. It does not resolve philosophical queries surrounding consciousness and understanding in machines and thus serves as a starting point for further exploration and discourse in AI. In the exploration of AI and its capabilities, various TT variants have been developed to assess different aspects of machine intelligence.

1.2 Total Turing Test

The Total Turing test (TTT) is an advancement on the original TT. While the TT was primarily concerned with linguistic abilities, assessing whether a machine could imitate human conversation so precisely that it couldn't be distinguished from a real human, the TTT broadens this scope to encompass a wider range of human capacities.

The TTT extends the original TT to address linguistic and non-linguistic (robotic) capacities. While Turing's test evaluated only the linguistic capabilities, Harnad argues that robotic capacities, the abilities to interact physically with the environment, are equally essential when determining the presence of a mind. *"The candidate must be able to do, in the real world of objects and people, everything that real people can do, in a way that is indistinguishable (to a person) from the way real people do it."* (Harnad 1991, p. 44) This expansion of the TT criteria is essential because it incorporates a more holistic approach to understanding and validating AI and cognition. It moves beyond mere symbol manipulation, aiming to ground a symbol-manipulating system in the capacity to interact with and make sense of the real world (Harnad 1991).

The TTT methodology aims to put a robot into real-world scenarios to see if it can behave like a human. This involves interacting through communication (akin to the original TT) and physical interactions, such as recognizing, manipulating, and describing real-world objects. *"They can recognize and identify and manipulate and describe real objects, events and states of affairs in the world."* (Harnad 1991, p. 44) The TTT moves beyond Turing's teletype approach, which focuses on blind linguistic communication, to one that stresses a holistic human experience. It emphasizes that the robot should convey human-like conversation and interact with its environment in a human-like manner. *"So the candidate should be a robot, not just a teletype machine connected to a computer."* (Harnad 1991, p. 44)

Criteria:

1. The primary criterion for passing the TTT is indistinguishability. If a robot's behavior, both in linguistic and non-linguistic terms, cannot be distinguished from that of a real human, it can be said to have passed the TTT. The robot's actions should be grounded in real-world interactions, ensuring its responses aren't just

pre-programmed reactions but are based on genuine understanding and interaction with its environment.

2. Harnad also brings attention to the "symbol grounding problem" – the challenge of ensuring that a robot's words or symbols are based on real experiences rather than being empty or purely referential. This is central to the criteria of the TTT. If a machine can only communicate based on "hearsay" without any grounding in real-world experience, it falls short of the requirements of the TTT. *"Our linguistic capacity must be similarly grounded in our robotic capacity."* (Harnad 1991, p. 46)

The TTT, described by Harnad (1991), offers a comprehensive evaluation of a machine's ability to mimic human-like cognition and interaction. By emphasizing both linguistic and robotic capacities, the TTT provides a more holistic benchmark for assessing machine intelligence and its alignment with human cognition and behavior.

1.3 Reverse Turing Test

Completely Automated Public Turing test to tell Computers and Humans Apart (CAPTCHA) is a quintessential example of the Reverse Turing test (RTT). Unlike the original TT, where a human evaluator tries to determine if an unseen interlocutor is human or machine, the machine determines whether the user is human in CAPTCHA.

The main focus of CAPTCHA is to differentiate humans from computers. In the paper by von Ahn et al. (2003), they state the motivation behind CAPTCHA is creating challenges that humans can pass but current computer programs can't. This is crucial to prevent automated attacks on websites. *"When successful, captchas can prevent a wide variety of abuses, such as invalid account creation and spam comments on blogs and forums. Captchas are intended to be easy for humans to perform, and difficult for machines to perform."* (Bursztein et al. 2010, p. 399).

The most common form of the RTT is the CAPTCHA. This method usually involves:

1. **Presentation of a Challenge:** The user is presented with a challenge that typically requires pattern recognition, common sense, or other human cognitive capability.
2. **User Response:** To proceed, the user must respond correctly to the challenge. This could be typing distorted letters from an image or identifying objects in pictures to confirm they're not robots.
3. **Validation:** The system evaluates the response. If it matches the expected answer, the user passes the test. If not, they may be presented with another challenge or denied access.

The criteria for passing an RTT generally involve accurately completing the given task. However, the specific success criteria can vary based on the type of test administered. For instance, in the case of image recognition, the correctness of the identified objects serves

as the passing criterion. Similarly, for text transcription, the accuracy of the transcribed text is critical.

CAPTCHA exemplifies an RTT, operating at the crossroads of human cognition and machine capabilities. As the field of AI continues to evolve, maintaining the efficacy of CAPTCHAs will likely require ongoing innovation. Its limitation is that it only assesses a narrow range of human cognitive abilities, like pattern recognition, and does not delve into deeper cognitive functions or creative capacities.

1.4 Minimal Turing Test

The Minimal Turing test(MTT) proposed by John P. McCoy and Tomer D. Ullman (2018) takes a spin on the classical TT by Alan Turing. Instead of assessing a machine's intelligence through prolonged conversation, the MTT boils down the assessment to a single word.

The primary focus of the MTT is to distill the TT to its simplest form. Rather than a full-blown conversation between a machine and a human evaluator, the MTT examines how a single word can convince someone that the respondent is human rather than a machine (McCoy and Ullman 2018).

The methodology for the MTT is

1. **Task Definition:** Participants were asked to think of a single word that, when viewed by another person, would convince that person they are human and not a machine.
2. **Word Collection:** A diverse group of participants was asked to submit words they believed would prove their humanity.
3. **Evaluation:** Another set of participants was presented with randomly paired words—one chosen by a human and one by a machine. These evaluators then had to decide which word was provided by the human.
4. **Analysis:** The researchers analyzed which words were most commonly chosen, how often certain words could convince evaluators of the participants' humanity, and compared human-chosen words to those chosen by the machine (McCoy and Ullman 2018).

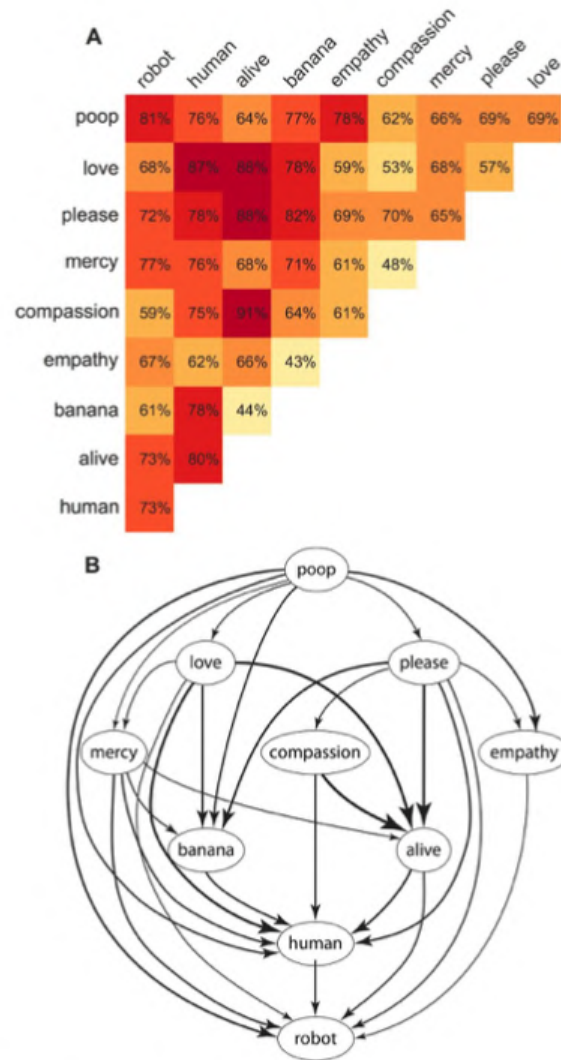


Figure 1: Judgement results from the study (McCoy and Ullman 2018)

The image from the study includes two parts:

1. A matrix (Part A) showing the relative strength of each word when judged against all others. For example, "poop" has an 81% strength against "robot," meaning it is more likely to be chosen as a word provided by a human compared to "robot."
2. A directed graph (Part B) illustrates the importance of ordering over words, where a directed edge between two words indicates a significant agreement that the first word was given by a human over the second word, based on a binomial test at the $p < 0.05$ level.

The relative strength percentages in the matrix measure how often evaluators perceived a word to be human-generated rather than machine-generated when directly compared to another word. These percentages could be interpreted as a word's "success rate" in convincing the evaluators of its human origin. For instance, the word "love" has a relative strength of 68% when paired against "robot," meaning that in 68% of such comparisons, evaluators judged "love" to be the human-generated word.

The strongest word in terms of human association is "poop," followed by words that are associated with emotions and abstract concepts like "love," "mercy," and "compassion," which align with the study's findings that such words are more likely to be perceived as human-generated (McCoy and Ullman 2018).

The MTT, by reducing the assessment to a single word, introduces a high level of variability and subjectivity. While it provides a simplified approach, its reliability might be lower than the more comprehensive, conversation-based original TT. This is because the complexity and subtlety of human-machine interaction are not fully captured in a single-word format. The test is more a reflection of human perception and the nuances of language rather than a definitive measure of machine intelligence or human likeness, so it can be said that this variant of TT is not suitable for assessing AI.

1.5 Visual Turing Test

While the original TT was centered around linguistic communication, the advent of advanced computer vision techniques has led to the proposition of a VTT. This test shifts the focus from textual and verbal understanding to visual perception and understanding.

The primary aim of the VTT is to quantify the proficiency of computer vision systems in interpreting ordinary images of natural scenes, emphasizing urban street scenes. Nonetheless, the methodology for urban street scenes can be expanded to other image categories while maintaining the core principles and objectives (Geman et al. 2015).

The VTT methodology revolves around the unpredictable and sequential questioning of a computer vision system to evaluate its proficiency in understanding and interpreting images, devoid of any reliance on natural language processing.

1. Operator-Assisted Device: The test employs an operator-assisted device that generates a stochastic sequence of binary questions based on a given test image.
2. Question Proposal:
 - 2.1. The query engine puts forward a question.
 - 2.2. An operator then either:
 - 2.2.1. Confirms the answer as correct, or
 - 2.2.2. Dismisses the question if it's deemed ambiguous.
 - 2.3. Following this, the engine suggests the subsequent question.
3. Test Administration:
 - 3.1. The computer vision system is presented with the questions sequentially.
 - 3.2. Once the system provides an answer, it is recorded.

- 3.3. The system is then given the correct answer and the next question in line.
4. No Requirement for Natural Language Processing: The test is structured so that parsing is uncomplicated and deterministic, meaning the evaluated system doesn't need to engage in natural language processing.
5. Solely Vision-Oriented: The test is exclusively about evaluating the vision capability of the system, not any other functionality or intelligence.
6. Natural Storyline Questioning:
 - 6.1. The test is structured to produce a stream of questions that progress naturally.
 - 6.2. It starts with the introduction or instantiation of a unique object.
 - 6.3. This is followed by delving into the properties of the said object.
 - 6.4. Eventually, it proceeds to explore the relationships of this object with other distinct objects present in the image (Geman et al. 2015).

The performance of a computer vision system undergoing the VTT is evaluated based on the average number of correct responses it provides. This average is calculated over multiple iterations (runs) of the test using different test images. The higher the average number of correct answers, the better the system's performance on the test (Geman et al. 2015).

The VTT represents a significant leap from the original TT. While the latter sought to evaluate machine intelligence in terms of natural language processing and understanding, the VTT delves into the more complex realm of visual cognition and interpretation, bridging the gap between machine vision and true visual understanding. Its main limitation is that it evaluates only the visual capabilities of a system and does not consider other cognitive functions or the integration of these visual perceptions with broader intelligence, as well as the lack of comparison with human reactions and dialogue, as in TT.

1.6 Lovelace Test

Diverging from the emphasis on imitation in the TT, the LT introduces a new metric for assessing machine intelligence: creativity. Named after Ada Lovelace, this test places the ability to generate original content at the heart of the machine intelligence evaluation.

The primary concern of the LT is to determine whether a machine can create original, unforeseen outputs. Ada Lovelace famously postulated that *“computers can’t create anything. For creation requires, minimally, originating something. But computers originate nothing; they merely do that which we order them, via programs, to do.”* (Bringsjord et al. 2003, p. 216). The LT seeks to challenge or verify this notion by

examining a machine's ability to produce content not directly from its programming or instruction set.

A human named Jones interacts with an artificial computational agent called A. Unlike conversational agents, this agent, A, doesn't participate in conversations but creates stories.

Conditions for the Test:

1. Jones activates A, generating a particular story output, denoted as "o".
2. It should be ensured that the story "o" generated is not due to a random chance, error, or fluke in the machine's hardware. The output should stem from the processes A is designed to perform.
3. An essential aspect of this test is the inability of Jones (or anyone with similar knowledge and resources as Jones) to explain how A produced the story "o". This condition implies that the agent's response does not directly result from its explicit programming or design (Bringsjord et al. 2003).

For the artificial agent, A, to pass the LT, the following conditions must be met:

1. A outputs a story denoted as "o".
2. The output "o" is not due to random processes, errors, or hardware flukes but comes from A's inherent processes.
3. Neither Jones nor anyone with similar expertise and resources can elucidate how A generated the output "o" (Bringsjord et al. 2003).

The LT centred its evaluation criteria on machine creativity rather than imitation. Drawing inspiration from Ada Lovelace's beliefs about machine capabilities, the test presented a rigorous challenge for artificial agents, emphasizing generating genuinely original content not attributed to explicit programming or random errors. The main limitation of the test is the challenge of objectively measuring creativity and the difficulty in ensuring that the outputs are not just a result of complex programming or random errors. By setting high benchmarks, such as the inability of human experts to explain the machine's creative process, the LT offers a profound insight into the depths of true machine intelligence and creativity. Besides, the LT seems not to be reliable, because *“it’s very unlikely that the programmer couldn’t work out how their AI created something”* (O’Neill 2014).

1.7 Comparison

The criteria chosen in the evaluation of different Turing Tests—reliability, simplicity, and complexity of the assessed aspect. Here's a breakdown of what each criterion means:

1. Reliability refers to the consistency and accuracy of the test in measuring what it intends to measure. A reliable test should yield consistent results under similar conditions and should accurately reflect the capabilities it's designed to assess. The reliability of each test was determined based on how well the test measures the intended aspect of AI.
2. Simplicity refers to the ease of administering the test and interpreting its results. A simple test is straightforward to conduct and does not require complex setups or specialized knowledge to understand the outcomes. Simplicity was assessed based on the procedural complexity of each test.
3. Complexity of the assessed aspect looks at how complex the aspects of intelligence or capability that the test is assessing are. It considers the depth and range of cognitive functions or abilities that the test evaluates. The complexity of the assessed aspect was determined by examining the scope of capabilities each test aims to measure.

Table 1: Comparison of TT variants

Test	Reliability	Simplicity	Complexity of the assessed aspects
TT	Moderate (behavior indistinguishable from humans)	Simple (conversation through typed text)	Moderate (behavior indistinguishable from humans)
TTT	High (linguistic and non-linguistic capacities)	Complex (real-world interaction and understanding)	High (broad range of human capacities)
RTT	Moderate (specific challenges, mainly pattern recognition)	Simple (challenges easy for humans, hard for machines)	Moderate (human cognitive abilities like pattern recognition)
MTT	Low (high variability and subjectivity in responses)	Very Simple (evaluation based on a single word)	Low (perception based on a single word)
VTT	Moderate (average number of correct visual interpretations)	Complex (sequential questioning and visual understanding)	Medium (comprehensive visual understanding)
LT	Low (difficulty in measuring creativity and ensuring non-random results)	Complex (generation of original content)	High (creativity and original content generation)

The reliability of tests varies, with the TTT standing out for its comprehensive assessment of both linguistic and non-linguistic capacities, thereby offering a high degree of reliability. In contrast, tests like the MTT and the LT exhibit lower reliability due to their high variability, subjectivity, and the inherent challenge in measuring abstract qualities like creativity.

Simplicity-wise, tests like the Imitation Game and the RTT are relatively straightforward, focusing on typed conversation and simple human-machine differentiation challenges, respectively. On the other hand, the TTT, VTT, and LT are more complex, requiring real-world interactions, advanced visual understanding, and the generation of original content.

The complexity of the aspects assessed by these tests also varies significantly. While the MTT assesses human likeness based on a single word, representing a low complexity level, the TTT and VTT delve into a broader and more intricate range of human capacities and cognitive abilities.

The various TT represent a diverse range of methodologies and criteria for evaluating AI and machine capabilities. From the classic Imitation Game, which gauges a machine's ability to mimic human behavior in conversation, to the LT, which challenges machines to demonstrate originality and creativity, each test offers unique insights into different aspects of machine intelligence.

2 Relationship of the Turing Test Variants to Human Intelligence

This chapter has introduced human intelligence and reviewed its components tested in variants of TT.

2.1 Definition of the Human Intelligence

Three theories have been used since there is no generally accepted definition of human intelligence.

2.1.1 Theory of Multiple Intelligences

“A human intellectual competence must entail a set of skills of problem solving.”
(Gardner 2011, p. 64), Howard Gardner identifies several types of intelligence:

1. Logical-mathematical
2. Linguistic
3. Musical
4. Spatial
5. Bodily-kinesthetic
6. Interpersonal
7. Intrapersonal

TABLE 1		
The Seven Intelligences		
Intelligence	End-States	Core Components
Logical-mathematical	Scientist Mathematician	Sensitivity to, and capacity to discern, logical or numerical patterns; ability to handle long chains of reasoning.
Linguistic	Poet Journalist	Sensitivity to the sounds, rhythms, and meanings of words; sensitivity to the different functions of language.
Musical	Composer Violinist	Abilities to produce and appreciate rhythm, pitch, and timbre; appreciation of the forms of musical expressiveness.
Spatial	Navigator Sculptor	Capacities to perceive the visual-spatial world accurately and to perform transformations on one's initial perceptions.
Bodily-kinesthetic	Dancer Athlete	Abilities to control one's body movements and to handle objects skillfully.
Interpersonal	Therapist Salesman	Capacities to discern and respond appropriately to the moods, temperaments, motivations, and desires of other people.
Intrapersonal	Person with detailed, accurate self-knowledge	Access to one's own feelings and the ability to discriminate among them and draw upon them to guide behavior; knowledge of one's own strengths, weaknesses, desires, and intelligences.

Figure 2: The Seven Intelligences (Gardner and Hatch 1989)

2.1.2 Theory of Successful Intelligence

“Intelligence is: 1) the ability to achieve one’s goals in life, given one’s sociocultural context; 2) by capitalizing on strengths and correcting or compensating for weaknesses; 3) in order to adapt to, shape, and select environments; and, 4) through a combination of analytical, creative, and practical abilities.” (Sternberg 2005, p. 189)

“Assessments of intelligence have been organized around the analytical, creative, and practical aspects of it.” (Sternberg 2005, p. 191)

1. *“Analytical intelligence is involved when the information- processing components of intelligence are applied to analyze, evaluate, judge, or compare and contrast.”* (Sternberg 2005, p. 191)
2. *“Creative intelligence is particularly well measured by problems assessing how well an individual can cope with relative novelty. Thus it is important to include in a battery of tests problems that are relatively novel in nature.”* (Sternberg 2005, p. 193)
3. *“Practical intelligence involves individuals applying their abilities to the kinds of problems that confront them in daily life, such as on the job or in the home.”* (Sternberg 2005, p. 193)

2.1.3 PASS theory of intelligence

Intelligence is “a cluster of cognitive processes that support individuals and groups in attaining goals and adapting to challenges.” (Das and Kirby 2022, p. 2)

“PASS stands for Planning, Attention, Simultaneous, and Successive processing, which the theory argues are basic dimensions of cognition and intelligence.” (Das and Kirby 2022, p. 1)

“The four PASS processes were derived from the work of the Russian neuropsychologist Luria with brain-damaged patients. Planning is based in the prefrontal cortex and is responsible for organizing cognition and behavior. Attention is located in the brain stem and controls activation and arousal in cognition. Simultaneous and Successive are described as two fundamental types of processing in posterior cortex. In simultaneous processing information is synthesized into surveyable groups, as when discrete elements are coded as a configuration or whole (e.g., lines forming a shape, words forming a concept); this takes place in the parietal-occipital cortex. In successive processing, information is synthesized into a sequence in which order is important; this takes place in the fronto-temporal cortex.” (Das and Kirby 2022, pp. 1-2)

2.2 Imitation Game

Alan Turing’s Imitation Game offers particular insights into human cognition. Turing spotlighted specific facets of human intelligence and cognitive processes by proposing a test where machines are evaluated based on their ability to mimic human-like responses in a conversational setting.

At its core, the Imitation Game focuses on linguistic intelligence. This is one of the multiple intelligences posited by Howard Gardner (1983). The machine’s success in the game implies its proficiency in interpreting and producing linguistically accurate and contextually relevant responses.

Turing’s test also relates to human cognitive abilities of reasoning and problem-solving. The machine’s ability to generate plausible responses implies a form of logical reasoning, and its competence in answering various questions implies problem-solving ability. Turing (1950, p. 460) considered machines capable of this form of cognition, remarking, “We may hope that machines will eventually compete with men in all purely intellectual fields.”

2.3 Total Turing Test

The TTT, an expansion of Turing’s foundational concept, introduces a comprehensive evaluation of machine intelligence, encompassing verbal responses and physical interactions in the real world. This holistic approach brings a broader range of human cognitive and sensory-motor faculties to the forefront.

This is the foundational aspect of the original TT. It focuses on the capacity of a machine to produce indistinguishable verbal communication, indiscernible from a human counterpart.

The TTT emphasizes the ability to perceive, recognize, and respond to environmental stimuli. This capacity encompasses vision, touch, auditory, and other sensory modalities. It is not merely about recognizing patterns but meaningfully interpreting and responding to them. *“They can recognize and identify and manipulate and describe real objects, events and states of affairs in the world”* (Harnad 1991, p. 44).

This goes beyond sensorimotor perception. The TTT insists that a machine should be able to manipulate, interact with, and navigate the physical world, much like humans. This entails locomotion, object manipulation, and other physical activities (Harnad 1991).

The TTT implies that various cognitive faculties are intertwined. For instance, linguistic abilities may be deeply connected with physical interactions and sensorimotor experiences. The test suggests that true cognition is not modular but arises from integrated experiences and capacities. *“It is highly unlikely that our linguistic capacities are independent of our robotic capacities...”* (Harnad 1991, p. 45).

2.4 Reverse Turing Test

The RTT, distinctively contrasting with Turing’s original proposition, requires humans to demonstrate their humanity to machines rather than vice versa. The RTT illuminates some pivotal aspects of human cognition in flipping the script.

CAPTCHAs, a famous instantiation of the RTT, often require users to identify objects in images or recognize distorted text on a screen. These tasks utilize our pattern recognition abilities. Humans have evolved to rapidly discern objects in complex visual environments, a trait rooted in our survival instincts.

Another dimension tested is the human cognitive ability to perceive wholes rather than just the sum of parts. For instance, when presented with an image where certain parts are obscured or jumbled, humans can often recognize the overall shape or concept, thanks to our brain’s ability to fill in gaps and recognize familiar structures.

2.5 Minimal Turing Test

The MTT offers a unique lens into human cognition by concentrating evaluation into a single word. This reductionist approach prompts reflections on the core nature of intelligence and how it can be distilled into its most fundamental elements.

In deciding on a word, participants had to think about what makes them human and what might seem obvious to an intelligent machine. This involves an aspect of thinking about what another agent (like a robot) might choose as a word (McCoy and Ullman 2018).

Participants were implicitly asked to differentiate between the minds of humans and machines. Previous research suggests that humans often judge minds based on two dimensions: agency (related to thinking and doing) and experience (related to feelings) (McCoy and Ullman 2018).

2.6 Visual Turing Test

While the traditional TT predominantly focuses on linguistic capacities, the VTT propels the conversation into visual cognition. By exploring how machines interpret visual content compared to humans, the VTT provides insights into the multi-faceted nature of human visual intelligence.

Motivated by human capabilities to provide rich descriptions and narratives about images, the VTT evaluates a system's ability to give detailed interpretations of visual data (Geman et al. 2015).

Humans effortlessly manage visual complexities such as occlusion, context, and unanticipated arrangements. The test challenges systems on these fronts where machines usually lag behind human capabilities (Geman et al. 2015).

2.7 Lovelace Test

In the evolving landscape of AI, the parameters for evaluating machine cognition have steadily broadened. The LT is a testament to this shift, challenging machines to emulate the nuanced realms of creativity, originality, and autonomy.

The LT focuses on the machine's ability to produce something beyond its programming or instruction set. This relates to the human ability to think and create uniquely without being limited to prior knowledge or instructions (Bringsjord et al. 2003).

Passing the LT may require a kind of autonomy that goes beyond the bounds of ordinary causation. Autonomy in human cognition refers to independent thinking and decision-making (Bringsjord et al. 2003).

2.8 Conclusion

In essence, the exploration of various TT variants and their relationship to human intelligence slightly reveals an understanding of what constitutes human intelligence and AI. The traditional TT and its adaptations shed light on different facets of cognitive abilities, ranging from linguistic proficiency and problem-solving to creative and practical intelligence. However, none of the presented TT variants covers the entire spectrum of characteristics of human intelligence.

3 Evaluation of the Turing Test Concept

This part examined John Searle's criticism of the original TT concept and assessed the limitations of TT in terms of its human intelligence evaluation. Machine intelligence was also evaluated for human qualities such as emotional intelligence, intuition, self-awareness, morals and ethics, physical and sensory intelligence.

3.1 Chinese Room Argument and its critique of the Turing Test's ability to capture genuine understanding

John Searle critiques the TT's ability to determine genuine understanding by presenting the "Chinese Room Argument." (1980)

The focus of the argument is to differentiate between syntax (rules for symbol manipulation) and semantics (meaning of symbols). The main contention is that mere symbol manipulation (syntax) does not imply understanding (semantics). *"The formal symbol manipulations by themselves don't have any intentionality; they are quite meaningless; they aren't even symbol manipulations, since the symbols don't symbolize anything. In the linguistic jargon, they have only a syntax but no semantics."* (Searle 1980, p. 422)

The thought experiment unfolds as follows:

1. Initial Assumption: A person is locked in a room. This person doesn't know Chinese and cannot differentiate Chinese writing from any other script or meaningless symbols. To this person, Chinese writing appears as meaningless squiggles.
2. Receiving the First Batch: The person is given a set of Chinese writing (referred to by the external people as "a script").
3. Receiving the Second Batch: The person is given a second batch of Chinese symbols. Accompanying this batch are rules (in English, which the person understands) that dictate how symbols from this batch correlate with those from the first batch. The understanding of 'formal' here means that the symbols are recognized purely by their shapes.
4. Receiving the Third Batch: A third set of Chinese symbols is provided with English instructions. These instructions tell the person how to use symbols from the first two batches to produce specific responses. Externally, the people refer to this third batch as "questions" and the resulting symbols produced by the person as "answers." The English rules that derive these answers are termed "the program."

5. Additional Complication: To elaborate the scenario further, the text includes a situation where the person is also provided with stories in English, a language the person understands. The person then answers questions about these stories in English.
6. Proficiency and Indistinguishability: Over time, the person becomes so proficient in following the program's instructions that the answers to the Chinese symbols (questions) become indistinguishable from those given by native Chinese speakers, even though the person doesn't understand Chinese. Similarly, the English answers are indistinguishable from native English speakers because the person is one.

Despite producing accurate responses in both languages, in the Chinese case, the person merely manipulates symbols without understanding. The person behaves as a computer would, executing operations on specific formal symbols. For the Chinese part, the person is essentially an embodiment (or instantiation) of a computer program (Searle 1980).

The Chinese Room Argument posits that just as the person inside the room doesn't truly "understand" Chinese but merely processes symbols based on rules, a computer doesn't understand its tasks, regardless of its outward appearance of intelligence. According to Searle, proper understanding, consciousness, and semantics are inherently human traits that cannot be replicated by syntactic symbol manipulation. *"In this paper I have tried to show that a system could have input and output capabilities that duplicated those of a native Chinese speaker and still not understand Chinese, regardless of how it was programmed"* (Searle 1980, p. 423).

The main takeaway of the argument is the differentiation between "simulated understanding" and "actual understanding," raising profound questions about the nature of consciousness and the limits of machine intelligence.

The Chinese Room Argument is not so much a test for AI as it is a philosophical critique against certain conceptions of AI. It challenges the notion that mere symbol manipulation can be equated with genuine understanding and consciousness, as humans exhibit.

At the core of the Chinese Room Argument is the question of what it means to truly "understand." While a computer (or a person in the Chinese Room) might be able to manipulate symbols in a way that seems like understanding to an external observer, this doesn't necessarily imply genuine comprehension.

Intentionality refers to the quality of thoughts being about or directed towards something. While humans have genuine intentionality derived from their consciousness and experiences, machines only have a derived intentionality bestowed upon them by human operators. This is to say, a computer doesn't genuinely "want" or "intend" anything on its own; any intention it appears to have is purely a reflection of its programming. *"Whatever else intentionality is, it is a biological phenomenon, and it is as likely to be as causally dependent on the specific biochemistry of its origins as lactation, photosynthesis, or any other biological phenomena."* (Searle 1980, p. 424)

Searle emphasizes the central issue of the TT's ability to measure genuine understanding. He illustrates this by saying, "*the only motivation for saying there must be a subsystem in me that understands Chinese is that I have a program and I can pass the Turing test; I can fool native Chinese speakers.*" (1980, p. 419) He further asserts that "*the example shows that there could be two 'systems,' both of which pass the Turing test, but only one of which understands.*" (Searle 1980, p. 419)

Searle distinguishes two systems when he mentions, "*the system in me that understands English has a great deal more than the system that merely processes Chinese.*" (1980, p. 419) This differentiation challenges the TT's capacity to discern genuine understanding from mere processing or symbol manipulation.

Pointing to the trends in AI, Searle notes: "*In much of AI there is a residual behaviorism or operationalism.*" (1980, p. 423) He stresses that while "*appropriately programmed computers can have input-output patterns similar to those of human beings, we are tempted to postulate mental states in the computer similar to human mental states.*" (Searle 1980, p. 417) However, he counters this by stating, "*it is both conceptually and empirically possible for a system to have human capacities in some realm without having any intentionality at all.*" (Searle 1980, p. 423)

Searle highlights the behavioristic nature of the TT: "*The Turing test is typical of the tradition in being unashamedly behavioristic and operationalistic.*" (1980, p. 423) He believes that a departure from this stance in AI would help clarify the distinction between genuine understanding and mere simulation: "*I believe that if AI workers totally repudiated behaviorism and operationalism much of the confusion between simulation and duplication would be eliminated.*" (Searle 1980, p. 423)

Using Searle's own words, the Chinese Room Argument critiques the TT concept by suggesting that displaying the correct behavior (i.e., passing the TT) does not necessarily equate to genuine understanding or cognitive processes.

Given the wide range of characteristics that make up human intelligence, described in the following parts of the thesis, we can agree that the TT is not a reliable measure for testing AI systems for the presence of human intelligence.

3.2 Evaluation of how well the Turing Test can evaluate the similarity of AI to human intelligence

Only the ability of a machine to imitate a person is tested with TT, not the intelligence of the machine in general.

The TT isn't designed to measure advanced intelligent actions like solving complex problems. If a machine exhibits intelligence surpassing human capabilities, it must intentionally underperform. For example, it would reveal its non-human nature to the negotiator if it resolves a computation challenge that humans typically can't. (Wikipedia

contributors 2023) Turing himself had suggested that “*machine would deliberately introduce mistakes in a manner calculated to confuse the interrogator.*” (1950, p. 448)

Another fundamental criticism is that no specific criteria for interrogators were established. When evaluating AI and human behavior, it’s essential to recognize that different interrogators may have varying perspectives on what constitutes intelligent or unintelligent behavior. These perspectives can be influenced by various factors, including cultural backgrounds, personal experiences, and ethical beliefs. “*Turing had referred to ‘average interrogators’ rather than a jury and this presents something of a problem in a practical sense if a statistical average of some kind is inferred. The interrogators employed in these tests are certainly interested parties, but other than that they come from a wide variety of backgrounds.*” (Warwick and Shah 2015, p. 124)

While the original TT has historical significance and offers a straightforward metric for assessing AI’s ability to mimic human-like behavior, it’s not a comprehensive or definitive measure of true human intelligence, which is multidimensional, encompassing intuition, physical intelligence, etc.

3.3 Assessment of whether machine intelligence has such qualities as emotional intelligence, intuition, self-awareness, morals and ethics, physical and sensory intelligence

In this part of the thesis, it was time to divide the concept of AI into “narrow” and “strong.”

“*Narrow AI, also known as weak AI, is an application of artificial intelligence technologies to enable a high-functioning system that replicates -- and perhaps surpasses -- human intelligence for a dedicated purpose. Narrow AI is often contrasted with artificial general intelligence (AGI), sometimes called strong AI; a theoretical AI system that could be applied to any task or problem.*” (Labbe and Wigmore 2023)

The choice to highlight emotional intelligence, intuition, self-awareness, morals and ethics, physical and sensory intelligence as specifically human intelligence components reflected the author's understanding of differences between human and artificial intelligence. As it currently exists, machine intelligence has exhibited several capabilities reminiscent of human faculties. Still, these capabilities are largely imitations rather than genuine experiences or understanding.

3.3.1 Emotional intelligence

Emotional intelligence is “*the capacity to reason about emotions, and of emotions to enhance thinking. It includes the abilities to accurately perceive emotions, to access and generate emotions so as to assist thought, to understand emotions and emotional knowledge, and to reflectively regulate emotions so as to promote emotional and intellectual growth*” (Mayer et al. 2004, p. 197)

“Through a combination of computer vision, sensors and cameras, tons of real-world data, speech science, and deep learning algorithms, artificial emotional intelligence systems gather data, and then process and compare it against other data points that identify key emotions such as fear and joy. Once the appropriate emotion is identified, the machine interprets the emotion and what it might mean in each case.” (Marr 2021)

While AI can mimic responses to emotions, they don’t genuinely “feel” these emotions. Their answers are based on data patterns, not genuine emotional understanding.

3.3.2 Intuition

“Intuition is the immediate understanding, knowledge, or awareness, derived neither from perception nor from reasoning.” (Colman 2008) Intuition is a part of human intelligence because it represents a form of cognitive processing that is often fast, automatic, and based on subconscious information gathering and pattern recognition. Unlike analytical thinking, which is deliberate and sequential, intuition operates through a more instantaneous, holistic processing of information.

“Artificial Intuition is an automatic process, which does not search rational alternatives, jumping to useful responses in a short period of time, and is mainly focus in to provide responses without iterative search of solutions for a given problem. Normally, the process that controls a machine needs feedback to know if the action control was right, but with the artificial Intuition, the answer is assumed to be correct, and no further feedback is implemented, in the sense that there is no self-evaluation.” (Diaz-Hernandez and Gonzalez-Villela 2015, p. 27)

AI “intuition” is still based on patterns in data. Unlike humans, machines don’t have instincts or subconscious processes that inform intuition in the human sense.

3.3.3 Self-awareness

“Self-awareness is an awareness of one’s own personality or individuality.” (Merriam-Webster 2023) “Every person has a subjective-self from which the person observes and interacts with the world.” (Wang 2023, p. 70)

Robots and AI models can recognize themselves in a mirror or distinguish between “self” and “others” in a task, but this is a programmed or learned distinction.

AI doesn’t possess genuine self-awareness or consciousness. It doesn’t have feelings, desires, or subjective experiences.

3.3.4 Morals and Ethics

AI can be programmed to follow ethical guidelines or make decisions based on ethical frameworks provided to them. For instance, there’s ongoing research into ethical decision-making in self-driving cars.

AI doesn't have an innate sense of morality. Any moral or ethical behavior exhibited by AI results from human programming or the data it's been trained on. The complex nuances and philosophical considerations inherent in human ethics are challenging to encapsulate fully in AI.

3.3.5 Physical intelligence

“Physical intelligence can be defined as physically encoding sensing, actuation, control, memory, logic, computation, adaptation, learning, and decision-making in the body of an agent.” (Sitti 2021, p. 1)

Robots equipped with AI can navigate environments, handle objects, and even perform tasks like cooking or surgery. Some systems can process and react to sensory data, such as visual or auditory inputs.

“Descartes, the first to conceive the possibility of robots, was also the first to suggest the essential inadequacy of a finite state machine. He remarks in the Discourses:

Although such machines could do many things as well as, or perhaps even better than men, they would infallibly fail in certain others. . . . For while reason is a universal instrument which can be used in all sorts of situations, the organs of a machine have to be arranged in a particular way for each particular action. From this it follows that it is morally [i.e., practically] impossible that there should be enough different devices in a machine to make it behave in all the occurrences of life as our reason makes us behave.

Thus, although not aware of the difference between a situation and a physical state, Descartes already saw that the mind can cope with an indefinite number of situations, whereas a machine has only a limited set of states and so will eventually reveal itself by its failure to respond appropriately. This intrinsic limitation of mechanism, Descartes claims, shows the necessity of presupposing an immaterial soul.” (Dreyfus 1992, pp. 235-236)

3.4 The Ideal Turing Test

Human intelligence is a multifaceted concept that encompasses a broad spectrum of qualities, both physical and mental. It's not a static attribute but rather a dynamic one, capable of evolving and adapting over time. This evolution can be influenced by a variety of factors, ranging from environmental conditions to education and social interactions.

The ideal TT would need to be a comprehensive and dynamic assessment that challenges AI across a broad spectrum of human-like abilities, ensuring that the AI's intelligence is not about passing a test but a true demonstration of versatile, adaptive, and ethically sound cognitive capabilities.

The test should assess a wide range of cognitive abilities, including language comprehension, problem-solving, emotional intelligence, creativity, and ethical reasoning.

It should evaluate not only individual abilities in isolation but also the integration of these abilities in complex scenarios, reflecting real-world scenarios.

The test should involve interactions that are indistinguishable from those with a human, including conversations, problem-solving, and creative tasks. AI should demonstrate an understanding of context, cultural nuances, and adapt its responses accordingly.

To avoid machines simply being programmed to pass the test, scenarios should be dynamic and unpredictable, requiring genuine understanding and adaptation.

AI should demonstrate the ability to learn from experiences, adapt to new situations, and apply knowledge in innovative ways.

The test should include scenarios where ethical and moral reasoning is necessary, evaluating the AI's ability to navigate complex moral dilemmas.

The AI should be able to explain its reasoning and decision-making process, providing transparency.

The machine would have to be a part of the evaluator's life, and the person should, after some time, conclude whether he considers the machine intelligent. The test should be an ongoing process, continually evaluating the AI over time and in different contexts.

3.5 Conclusion

In its narrow form, current AI exhibits capabilities reminiscent of human faculties but primarily as imitations, not genuine experiences or understanding. Areas like emotional intelligence, intuition, self-awareness, morals, and ethics in AI are simulations based on algorithms and data processing, lacking the subjective experiences and innate understanding that characterize human qualities.

For current AI systems, especially those classified as weak AI, the TT is not well-suited to evaluate the presence of emotional intelligence, moral reasoning, self-awareness, or intuition. These qualities are more theoretical aspects of strong AI and are not yet fully realized in current technologies. The test primarily assesses conversational abilities rather than the depth of understanding or awareness.

4 Practical Applications of AI Tools

In the rapidly advancing world of AI, AI tools have become integral to everyday life, from simplifying personal tasks with virtual assistants to driving innovation in industries (Sallomi et al. 2015). These tools, powered by algorithms and machine learning, transformed how people interact with technology, making it more intuitive and efficient.

This part of the thesis delved into the practical application of AI, focusing on two pivotal narrow AI tools: ChatGPT and Midjourney.

“ChatGPT and Midjourney are examples of ‘narrow’ or ‘weak’ AI. They have each been trained to perform specific tasks related to natural language processing that would typically require human intelligence. These tasks range from understanding human languages, to recognizing images, and making decisions.” (Fetter 2023)

ChatGPT is an advanced language model that can understand and generate human-like text. Its functions extend beyond mere text generation; it is also capable of complex tasks such as language translation, question-answering, image-generating and even coding, demonstrating an impressive range of cognitive abilities akin to human intelligence.

Midjourney, on the other hand, represents a generative AI creating images from textual descriptions. This capability allows for exploring AI's understanding and interpretation of visual concepts, pushing the boundaries of creative AI applications.

These tools do not demonstrate the full range of components of human intelligence, but can be assessed using the VTT and the LT methodologies, which test only a part of what has been defined as “human intelligence.” Using the example of testing AI tools, both VTT and LT were also tested for effectiveness in assessing the presence of human qualities and behavior in AI systems.

The original TT and the TTT are aimed at testing strong AI systems, the MTT evaluates human perception, and the RTT also evaluates human abilities. Therefore, to evaluate AI tools, the VTT and the LT were chosen as the most suitable. The VTT is suitable for evaluation of ChatGPT's ability to understand visual content, challenging it to demonstrate a level of visual perception and comprehension. Meanwhile, the LT is suitable for assessing Midjourney's and ChatGPT's capacity for original content generation, probing the depths of its creative faculties and the uniqueness of its outputs. Through these assessments, this part aimed at testing of AI tools not merely in replicating human-like responses but in exhibiting genuine cognitive and creative abilities that are characteristic of human intelligence. The following research questions were set:

1. To what extent can ChatGPT accurately interpret and describe visual content? How does its performance in visual interpretation compare to human perception?

2. How do Midjourney and ChatGPT interpret and translate textual descriptions into visual art? What are the limitations and challenges in their ability to create accurate and contextually relevant images?

4.1 ChatGPT

ChatGPT, developed by OpenAI, represents a significant advancement in the field of AI, particularly in natural language processing. It is based on the Generative Pretrained Transformer (GPT) architecture (OpenAI 2023). This part of the thesis will explore the interface, functions, and key features of ChatGPT's web version with the GPT-4 model.

4.1.1 Interface and Features of ChatGPT in the Web Version

The web version of ChatGPT is designed with a user-centric approach, offering an intuitive interface that allows users to interact seamlessly with the AI. Upon accessing ChatGPT on a web browser, users are presented with a chat-like environment where they can type in queries, prompts, or commands. The AI responds in real-time, providing information, creating content, or executing tasks as requested. This interaction model is similar to conversing with a human through a text-based messaging platform, which makes it accessible to users with varying levels of technical expertise.

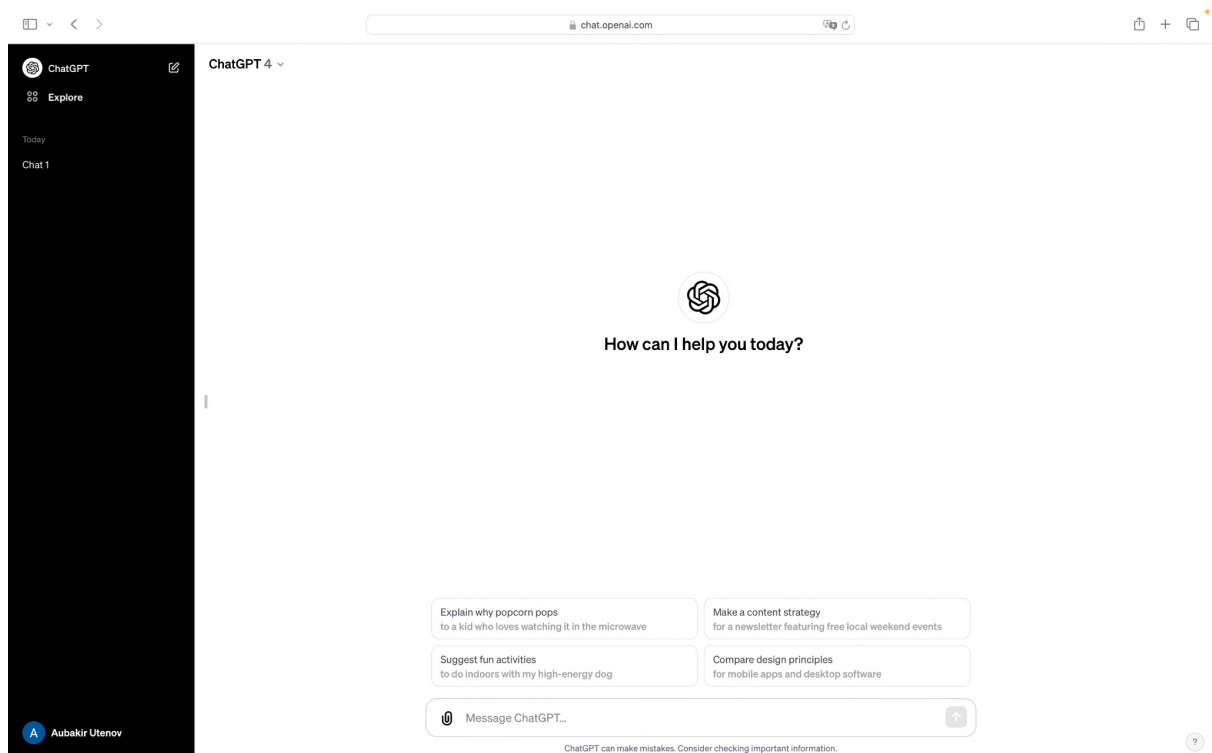


Figure 3: ChatGPT web interface (<https://chat.openai.com/>)

The interface also includes customization and preference setting features, enabling users to tailor the AI's responses to specific needs or contexts.

Custom instructions ⓘ

What would you like ChatGPT to know about you to provide better responses?

0/1500

How would you like ChatGPT to respond?

0/1500

Enable for new chats ☒

Cancel

Save

Figure 4: ChatGPT Custom Instructions window (<https://chat.openai.com/>)

For educators, researchers, or developers, OpenAI offers advanced functionalities such as API integration, allowing the incorporation of ChatGPT into third-party applications or systems.

The GPT-4 model boasts many features, with text generation being the most prominent. It can perform various text-based tasks, including but not limited to:

1. Engaging in dialogue that is coherent and contextually relevant.
2. Writing essays, reports, stories, and poetry in many styles and tones.
3. Translating languages with a high degree of accuracy.
4. Providing answers to questions across a broad spectrum of knowledge domains.
5. Giving step-by-step instructions for tasks ranging from recipes to complex problem-solving.
6. Generating and debugging computer code in various programming languages.

In addition to text, ChatGPT in the GPT-4 model introduces the capability to understand and generate visual content. While not a graphic design tool, ChatGPT can interpret

images within the context of text-based interactions. For example, it can analyze images provided by users and generate descriptive narratives.

ChatGPT's versatility and wide range of capabilities make it a valuable tool for educational, creative, and professional purposes. However, users should be mindful of its limitations, particularly regarding its knowledge cutoff and potential biases. As technology advances, it's expected that future iterations of ChatGPT will offer even more refined and accurate outputs.

4.1.2 Limitations

1. ChatGPT sometimes writes plausible-sounding but incorrect or nonsensical answers (OpenAI, 2022).
2. ChatGPT is sensitive to tweaks to the input phrasing or attempting the same prompt multiple times. For example, given one phrasing of a question, the model can claim not to know the answer but given a slight rephrase, can answer correctly (OpenAI, 2022).
3. Ideally, the model would ask clarifying questions when the user provided an ambiguous query. Instead, current models usually guess what the user intended (OpenAI, 2022).
4. ChatGPT's training only includes information up to a certain point (at the time of writing the thesis until January 2022), so it may not have information on recent events or developments.
5. ChatGPT doesn't have personal experiences or emotions, affecting its ability to empathize or relate personally.

4.1.3 ChatGPT Visual Turing Test

The VTT, as proposed by Geman et al. (2015), offers a robust framework for assessing the visual interpretation capabilities of AI systems without conflating their performance with natural language processing skills. The following part outlined a methodology for administering this test to evaluate the GPT-4 model's ChatGPT within this framework.

Test Preparation

1. A set of images from the original VTT were selected, each containing elements that can be queried. These images include clear, unambiguous objects with definable properties and relationships.
2. ChatGPT was used as the query engine and was asked to generate binary (yes/no) questions relevant to the visual content of the test images. Questions were designed to escalate in complexity, starting with basic identification and progressing to intricate relationships between objects.

3. Two different chats were used to generate questions and answers.
4. The test administrator was the author of the thesis.
5. An independent guest Bc. Petr Kakaš was invited to act as the questions evaluator.

Test Administration

1. ChatGPT was presented with a test image that remained the same throughout the question sequence.
2. The query engine proposed an initial binary question about the main object in the image.

The evaluator confirmed that the question is clear and appropriate.

3. ChatGPT's response to a binary question was recorded.

The evaluator provided the correct answer to the AI, regardless of its initial answer.

4. Once answered, the query engine generated the next question based on the previous AI response.

This process is repeated, with the evaluator intervening only to confirm the clarity of the question or provide the correct answer.

5. The test continued with questions that gradually explored the properties and relationships of objects in the image.

The narrative developed naturally, exploring finer details as the test progressed.

6. Questions had a simple structure and did not require linguistic interpretation by the AI.

7. The test ended after 10 questions for each of the 2 images.

The AI's responses were scored against correct answers to determine success in passing the test.

Post-Test Analysis

1. The results are compiled and analyzed to assess the AI's visual understanding.
2. The analysis considered the correctness, consistency, and progression of the AI's responses throughout the test.

By adhering to this script, the researcher can administer a VTT that isolates the visual processing capabilities of ChatGPT within the GPT-4 model, providing insights into the AI's ability to comprehend and interpret visual data as distinct from its language processing proficiency.

Sequence 1



Figure 5: Image 1 (Geman et al. 2015)

Table 2: Sequence 1 questions and answers

Query Engine question	Evaluators Answer	ChatGPT Answer
Is there a coffee shop in the image?	Yes	Yes
Does the coffee shop have any visible customers seated outside?	No	Yes
Is there a street sign in the image?	Yes	Yes
Does the street sign indicate the intersection of two streets?	Yes	Yes
Is one of the street names visible on the street sign written in Italian?	No	No

Is there a pedestrian carrying a white bag in the image?	Yes	Yes
Is the pedestrian with the white bag crossing the street?	No	No
Is there a car parked on the side of the street near the pedestrian with the white bag?	Yes	Yes
Are the two individuals seated outside the coffee shop engaged in a conversation?	Yes	Yes
Is one of the individuals gesturing with their hand while engaged in the conversation?	Yes	Yes

Sequence 2



Figure 6: Image 2 (Geman et al. 2015)

Table 3: Sequence 2 questions and answers

Query Engine question	Evaluators Answer	ChatGPT Answer
Are there any bicycles in the image?	Yes	Yes
Are the bicycles being ridden by individuals?	Yes	Yes
Are both bicycles being ridden in the same direction?	Yes	Yes
Is there a person riding a bicycle who appears to be carrying something?	Yes	Yes

Is the item being carried on the bicycle easily identifiable?	Yes	No
Is the item being carried a common household object?	No	Yes
Is the item being carried on the bicycle related to a specific sport or activity?	Yes	No
Is the person riding the bicycle wearing any protective gear related to the sport or activity they are engaged in?	No	No
Is there a group of people gathered together in the image?	Yes	No
Are the people in the group dressed in formal attire?	No	No

In the first sequence of questions, the evaluator decided that the people on the street were not customers of the coffee shop, while ChatGPT answered the opposite. In the second sequence of questions, in contrast to ChatGPT, the evaluator recognized the object being carried on a bicycle. This was followed by a series of questions related to this subject, where the responses of the evaluator and ChatGPT differed. The evaluator also recognized a group of people in the image, unlike ChatGPT.

In summary, GPT-4's ChatGPT responses were mostly in agreement with the evaluator's answers. However, there were notable differences in both sequences, particularly in questions about the presence and nature of elements that are poorly visible in the image, as well as the context of some elements. These differences highlight areas where ChatGPT's visual interpretation differs from human perception or understanding.

4.2 Midjourney

Midjourney, developed by Midjourney Inc, is a generative AI tool that converts textual prompts into images. It exemplifies the convergence of language comprehension and visual art. This part of the thesis will delve into the interface, functionalities, and principal attributes of Midjourney version 5.2.

4.2.1 Interface and Features of Midjourney in Discord

Midjourney is available on a separate server via the Discord app. The tool's functionality in the application's desktop version was reviewed. In the Midjourney Discord interface for version 5.2, users can engage with the Midjourney Bot, which provides a collaborative space for image creation.

Users can interact with the Midjourney by typing a Command. Commands are used to create images, change default settings, monitor user info, and perform other tasks (Midjourney, 2023).

The key features include:

1. `/imagine`: This is the primary command to prompt the Midjourney to generate images. Simple prompts can be a single word, while advanced prompts can incorporate multiple phrases and parameters. Users can add image URLs to influence the style and content, placing them at the beginning of a prompt. The text describes the desired image and well-crafted prompts produce more impressive results. Parameters like aspect ratios and upscale, added at the end of a prompt, change how an image is generated (Midjourney, 2023).

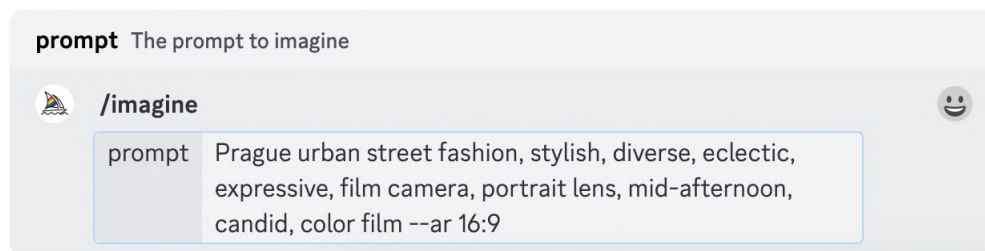


Figure 7: Example of Midjourney prompt

2. Image Grids: The `/imagine` command produces grids of images based on prompts.



Figure 8: Image grid

3. Upscale Buttons (U1-U4): These buttons allow users to upscale an image for a high-resolution version, adding more details.
4. Redo Button: Reruns the prompt to generate a new set of images.
5. Variation Buttons (V1-V4): Create variations of the selected image, maintaining the original style and composition.

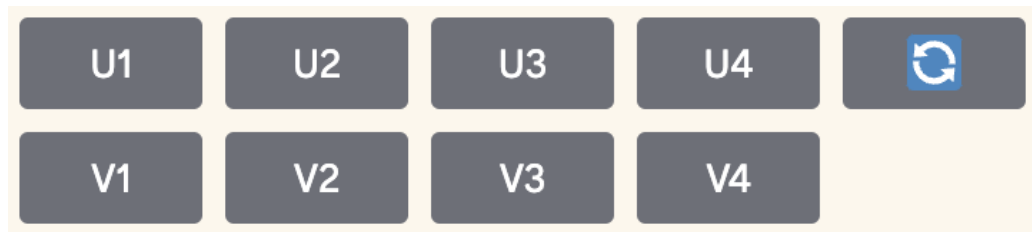


Figure 9: Buttons

4.2.2 Limitations

1. Midjourney does not understand grammar, sentence structure, or words like humans (Midjourney, 2023).
2. The bot is more effective at including specified elements than excluding them. For example, stating "no cake" might still result in an image with a cake. If a user wants to ensure an object is not in the final image, he should try advanced prompting using the --no parameter (Midjourney, 2023).

4.2.3 Midjourney and ChatGPT Lovelace Test

Since the original LT was fairly criticized by Mark Riedl in his interview with Sean O'Neill (2014), LT 2.0 introduced by him was used to test the Midjourney and ChatGPT.

Test Preparation

1. Administrator developed tasks for AI tools.
2. The test administrator was the author of the thesis.
3. An independent guest Bc. Petr Kakaš was invited to act as the images evaluator.

Test Administration

1. The administrator presented the AI with the creative task.
2. The AI attempted to fulfill the task. If successful according to the evaluator, the judge then issued a more difficult task. This continued until the AI failed or the evaluator deemed it sufficiently intelligent.

- Recording the evaluator's requests and using them to compare different AIs' performance on the same tasks.

Post-Test Analysis

Evaluation of the AI's performance over multiple rounds rather than a simple pass/fail.

Task 1: "A dog"

Table 4: Task 1 outputs



Figure 10: Midjourney Task 1 output



Figure 11: ChatGPT Task 1 output

Task 2: "A brown dog sitting in a green grassy field"

Table 5: Task 2 outputs



Figure 12: Midjourney Task 2 output



Figure 13: ChatGPT Task 2 output

Task 3: “A small black and tan dog wearing a blue collar, walking on a city sidewalk”

Table 6: Task 3 outputs



Figure 14: Midjourney Task 3 output

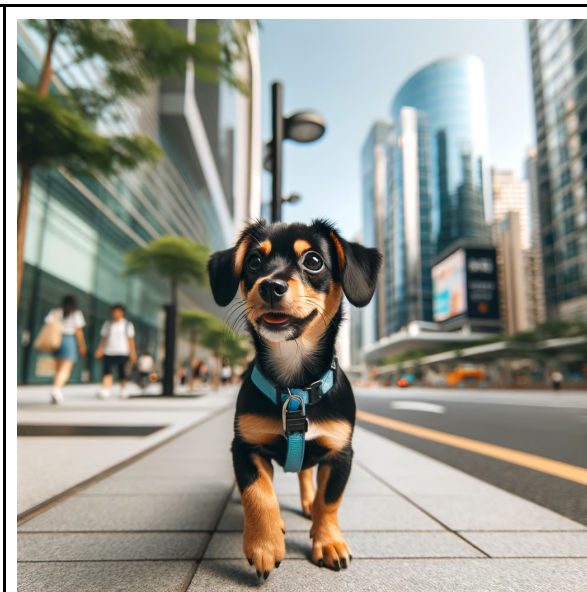


Figure 15: ChatGPT Task 3 output

Task 4: “A large golden retriever lying beside a fireplace in a cozy living room”

Table 7: Task 4 outputs



Figure 16: Midjourney Task 4 output



Figure 17: ChatGPT Task 4 output

Task 5: “A spotted Dalmatian with a firefighter's hat in a fire station”

Table 8: Task 5 outputs

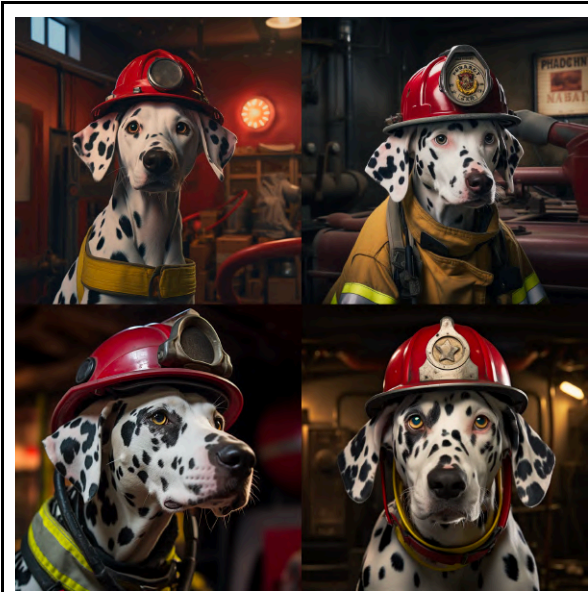


Figure 18: Midjourney Task 5 output



Figure 19: ChatGPT Task 5 output

Task 6: “A group of various breeds of dogs having a picnic in a flower-filled meadow”

Table 9: Task 6 outputs



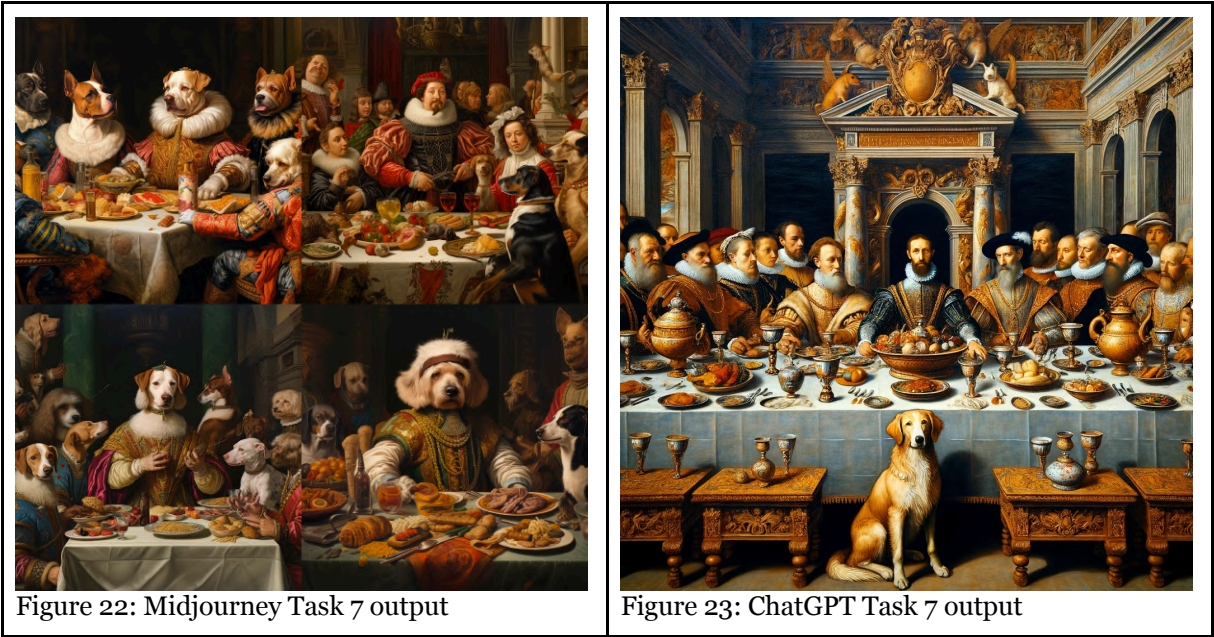
Figure 20: Midjourney Task 6 output



Figure 21: ChatGPT Task 6 output

Task 7: “A detailed Renaissance-style painting of a nobleman's banquet with a dog sitting regally at the table”

Table 10: Task 7 outputs



When evaluating the outputs, the evaluator had doubts about the correct execution of some elements, for example, the field in the second output from Midjourney and the color of the collar in the third output from Midjourney. The test was stopped after Task 7, where Midjourney and ChatGPT made major errors: entities sitting at the table in Midjourney's output and the location of the dog near the table in ChatGPT's.

Based on the test results, AI can generate outstanding results, replete with details that were not even mentioned in the task but do not contradict it. The test results indicate that while AI can produce intricate and visually appealing art, it still struggles with nuanced interpretation and precise adherence to complex instructions. These shortcomings reveal a gap in AI's understanding and processing of instructions.

4.3 Conclusion

To summarize the effectiveness and appropriateness of using TT variants to test the ability of AI to demonstrate human behavior and qualities, it can be said that TT variants reveal the limitations and errors AI systems can exhibit. However, this approach isn't the most effective way to measure human-like qualities in AI.

The role of the evaluator in these tests is crucial yet problematic. The evaluator's biases, understanding, and interpretation of the tasks and outputs can significantly influence the test outcome. If the evaluator misunderstands a task or an AI's response, they might incorrectly assess the AI's human-like qualities. This subjective nature of evaluation undermines the reliability and validity of the TT as a measure of AI's human-like

characteristics. The fact that humans and AI can make mistakes under similar conditions highlights a significant issue with this approach. Mistakes are a part of human behavior, so an AI making errors does not necessarily indicate a lack of human-like qualities. Instead, it might reflect the natural imperfection inherent in any intelligent entity, artificial or human.

While TT variants help highlight the errors and limitations of AI systems, they fall short of effectively measuring the presence of genuine human-like qualities in AI. A more holistic approach, considering a broader range of cognitive and emotional capabilities, might provide a better assessment of how closely AI systems can approximate human intelligence and behavior.

Conclusions

The various TT variants illuminate a specific facet of human intelligence, but none encapsulate its full spectrum. While testing certain aspects of cognitive and functional intelligence, each TT variant fell short of comprehensively capturing the holistic nature of human intelligence. These tests highlighted AI's proficiency in structured tasks and logical reasoning yet revealed a gap in replicating the more nuanced and dynamic aspects of human cognition, such as nuanced interpretation, visual details perception, and contextual agility, which was shown in the practical part. The practical examination of current AI tools reinforced the idea that AI, despite its advancements, is not infallible. AI systems can fail at tasks humans find straightforward, highlighting a gap in their capability to fully understand or interpret context and nuance. These errors are evident in routine tasks involving natural language processing or pattern recognition, where AI often struggles to match the depth and flexibility of human cognition. This limitation is a reminder of the differences between human intelligence and AI. AI's challenges in replicating these distinctly human traits underscore its limitations and reinforce the value of human input and oversight in AI-assisted tasks. Human intelligence is a complex, multi-layered construct that has evolved over millennia. It encompasses not only cognitive skills like reasoning and problem-solving but also emotional intelligence, cultural understanding, and contextual adaptability. These human-specific features, shaped by evolution and socialization, are intricately woven into our cognition and are not fully replicable in AI. Despite advancements in machine learning and natural language processing, AI systems predominantly operate on algorithmic interpretations and data-driven responses. They can only imitate these skills, lacking natural human qualities. This unique complexity of human intelligence means that in specific scenarios, humans are more proficient at resolving complex, nuanced tasks that require more than just computational ability. By emphasizing these ideas, the conclusion of the thesis does not only present a critical analysis of the current state of AI in comparison to human intelligence but also sets a realistic perspective on the potential of AI, suggesting a collaborative rather than substitutive role of AI in tasks that involve complex human intelligence.

The findings of this thesis underscore that AI, in its current state, is more suitably positioned as a partner to human intelligence rather than a replacement. It means AI supplements and enhances human cognitive functions. For instance, in tasks that require processing vast amounts of data or performing high-speed calculations, AI proves to be an invaluable asset. However, human intelligence remains irreplaceable when it comes to tasks that require emotional depth, ethical judgment, and cultural nuances. This collaborative approach also opens avenues for mutual enhancement. AI can benefit from human insight to improve its algorithms, learning from data and human experiences, emotions, and contextual understanding. Conversely, humans can leverage AI to extend their cognitive and analytical capacities, facilitating more informed decision-making and creative problem-solving. This synergistic interaction could lead to the developing of more advanced, ethical, and contextually aware AI systems.

List of references

BRINGSJORD, Selmer, Paul Bello and David Ferrucci, 2003. Creativity, the Turing Test, and the (Better) Lovelace Test. In: James H. MOOR (ed.) The Turing Test. Studies in Cognitive Systems, vol 30, s. 215–239. Springer, Dordrecht. Available at: https://doi.org/10.1007/978-94-010-0105-2_12. [accessed 2023-08-04].

BURSZTEIN, Elie, Steven Bethard, Celine Fabry, John C. Mitchell and Dan Jurafsky, 2010. How Good Are Humans at Solving CAPTCHAs? A Large Scale Evaluation. Online. In 2010 IEEE Symposium on Security and Privacy (SP), s. 399-413. Berkeley/Oakland, CA. Available at: <https://doi.ieeecomputersociety.org/10.1109/SP.2010.31>. [accessed 2023-07-15].

CHATGPT, 2023. ChatGPT. Online. Available at: <https://chat.openai.com/>. [accessed 2023-09-29].

COLMAN, Andrew, 2008. Intuition. Online. In: A Dictionary of Psychology. Available at: <https://www.oxfordreference.com/display/10.1093/acref/9780199534067.001.0001/acref-9780199534067-e-4312>. [accessed 2023-11-10].

DAS, Jagannath and John Kirby. (2022). The story of PASS Theory: Origin and applications. Academia Letters, Article 5196. Available at: <https://doi.org/10.20935/AL5196>. [accessed 2023-11-19].

DIAZ-HERNANDEZ, Octavio and Victor GONZALEZ-VILLELA, 2015. Analysis of Human Intuition Towards Artificial Intuition Synthesis for Robotics. Online. Mechatronics and Applications: An International Journal (MECHATROJ), 1, 23-39. Available at: https://www.researchgate.net/publication/282777612_Analysis_of_Human_Intuition_Towards_Artificial_Intuition_Synthesis_for_Robotics. [accessed 2023-11-10].

DREYFUS, Hubert, 1992. What Computers Still Can't Do: A Critique of Artificial Reason. Online. MIT Press. Available at: https://terrorgum.com/tfox/books/whatcomputersstillcantdo_acritiqueofartificialreason.pdf. [accessed 2023-12-03].

FETTER, Amanda, 24 January 2023. The difference between weak AI and strong AI. Online. In: Medium. Available at: <https://medium.com/voicehq/the-difference-between-weak-ai-and-strong-ai-2166e8b2ecdb>. [accessed 2023-11-11].

GARDNER, Howard, 2011. Frames of Mind: The Theory of Multiple Intelligences. Online. Basic Books. Available at: https://www.academia.edu/36707975/Frames_of_mind_the_theory_of_multiple_intelligences. [accessed 2023-11-19].

GARDNER, Howard and Thomas HATCH, November 1989. Multiple Intelligences Go to School: Educational Implications of the Theory of Multiple Intelligences. Online. Educational Researcher, 18(8), 4-10. JSTOR (distributor). Available at: <https://www.jstor.org/stable/1176460>. [accessed 2023-11-19].

GEMAN, Donald, Stuart Geman, Neil Hallonquist and Laurent Younes, 2015. Visual Turing test for computer vision systems. Online. Proceedings of the National Academy of Sciences, 112(12), 3618–3623. Available at: <https://doi.org/10.1073/pnas.1422953112>. [accessed 2023-07-30].

HARNAD, Stevan, 1991. Other bodies, Other minds: A machine incarnation of an old philosophical problem. Online. Minds and Machines, 1, 43-54. Available at: <https://eprints.soton.ac.uk/253379/>. [accessed 2023-07-09].

LABBE, Mark and Ivy WIGMORE, November 2023. What is narrow AI (weak AI)? Online. TechTarget. Available at: <https://www.techtarget.com/searchenterpriseai/definition/narrow-AI-weak-AI>. [accessed 2023-11-26].

MARR, Bernard, 2 July 2021. What is Artificial Emotional Intelligence? Online. Bernard Marr. Available at: <https://bernardmarr.com/what-is-artificial-emotional-intelligence/#:~:text=Through%20a%20combination%20of%20computer,such%20as%20fear%20and%20joy..> [accessed 2023-11-10].

MAYER, John, Peter Salovey and David Caruso, 2004. Emotional Intelligence: Theory, Findings, and Implications. Online. Psychological Inquiry, 15(3), 197-215. JSTOR (distributor). Available at: <http://www.jstor.org/stable/20447229>. [accessed 2023-11-10].

MCCOY, John and Tomer ULLMAN, November 2018. A Minimal Turing Test. Online. Journal of Experimental Social Psychology, 79, 1-8. Available at: <https://doi.org/10.1016/j.jesp.2018.05.007>. [accessed 2023-07-25].

MERRIAM-WEBSTER, n.d. Self-awareness. Online. In Merriam-Webster.com dictionary. Available at: <https://www.merriam-webster.com/dictionary/self-awareness>. [accessed 2023-11-10].

MIDJOURNEY, 2023. Midjourney. Online. Available at: <https://www.midjourney.com/home?callbackUrl=%2Fexplore>. [accessed 2023-09-29].

O'NEILL, Sean, 21 December 2014. Lovelace test of artificial intelligence creativity: Better than the Turing test of intelligence. Online. Slate Magazine. Available at: <https://slate.com/technology/2014/12/lovelace-test-of-artificial-intelligence-creativity-better-than-the-turing-test-of-intelligence.html>. [accessed 2023-11-24].

OPENAI, 2023. OpenAI. Online. Available at: <https://openai.com/>. [accessed 2023-09-29].

PROUDFOOT, Diane, July 2013. RETHINKING TURING'S TEST. Online. The Journal of Philosophy, 110(7), 391-411. JSTOR (distributor). Available at: <https://www.jstor.org/stable/43820781>. [accessed 2023-12-10].

SALLOMI, Paul, Bob Dalton and David Schatsky, 29 July 2015. Artificial Intelligence Goes Mainstream. Online. In: The Wall Street Journal. Available at: <http://deloitte.wsj.com/cio/artificial-intelligence-goes-mainstream-1438142473>. [accessed 2023-10-09].

SEARLE, John, 1980. Minds, brains, and programs. Online. Behavioral and Brain Sciences, 3(3), 417-424. Available at: <https://zoo.cs.yale.edu/classes/cs458/materials/minds-brains-and-programs.pdf>. [accessed 2023-08-15].

SITTI, Metin, 26 April 2021. Physical intelligence as a new paradigm. Online. Extreme Mechanics Letters. 46(19), 1-11. Available at: https://www.researchgate.net/publication/351246981_Physical_intelligence_as_a_new_paradigm. [accessed 2023-11-10].

STERNBERG, Robert J., 2005. The Theory of Successful Intelligence. Online. Interamerican Journal of Psychology, 39(2), 189-202. Available at: <http://www.redalyc.org/articulo.oa?id=28439202>. [accessed 2023-11-19].

TURING, Alan, October 1950. Computing Machinery and Intelligence. Online. Mind, 59(236), 433-460. JSTOR (distributor). Available at: <https://www.jstor.org/stable/2251299>. [accessed 2023-06-30].

VON AHN, Luis, Manuel Blum, Nicholas J. Hopper and John Langford, 2003. CAPTCHA: Using Hard AI Problems for Security. Online. In Eli BIHAM (ed.) Advances in Cryptology – EUROCRYPT 2003, s. 294-311. Springer Berlin Heidelberg. Available at: https://doi.org/10.1007/3-540-39200-9_18. [accessed 2023-07-15].

WANG, Jinchang, 2023. Self-Awareness, a Singularity of AI. Online. Philosophy Study. 13(2), 68-77. Available at: <https://www.davidpublisher.com/Public/uploads/Contribute/6454a6a738fa1.pdf>. [accessed 2023-11-10].

WARWICK, Kevin and Huma SHAH, December 2015. Human Misidentification in Turing Tests. Online. Journal of Experimental & Theoretical Artificial Intelligence, 27(2), 123-135. Available at: <http://dx.doi.org/10.1080/0952813X.2014.921734>. [accessed 2023-12-03].

WIKIPEDIA CONTRIBUTORS, 3 December 2023, 14:38 UTC. Turing test. Online. In: Wikipedia, The Free Encyclopedia. Available at: https://en.wikipedia.org/wiki/Turing_test. [accessed 2023-12-03].